Comprehensive Benchmark of Loss Functions for ECG Reconstruction

Antigravity Agent

July 31, 2026

1 Clinical Ablation Study

To rigorously assess the impact of each loss component on clinical fidelity, we performed an exhaustive combinatorial ablation study. The factorial matrix evaluated 28 distinct model configurations. We analyzed the reconstructions across 334 records from the Zhejiang PVC database, focusing on specific temporal morphological boundaries (P, QRS, T wave onsets and offsets) and lead-specific amplitudes.

1.1 Spatial Regularization via VCG and Einthoven Consistency

Our primary finding is the regularization effect of combining Vectorcardiogram (VCG) spatial projection with Einthoven lead consistency. The baseline Mean Squared Error (MSE) model exhibits a Mean Absolute Error (MAE) of 15.6 $\pm$ 2.4 μV on Lead V6. However, introducing the VCG and Einthoven constraints (Mask `1001014`) significantly reduces the V6 MAE to 2.0 $\pm$ 0.3 μV.

Clinically, Lead V6 is a unipolar precordial lead that observes the lateral wall of the left ventricle. Standard temporal models often allow this lead to drift independently. By enforcing the 3D spatial dipole via the VCG projection loss, the model is geometrically constrained. The Einthoven consistency further anchors the limb leads, providing a mathematical scaffold that prevents individual chest leads from drifting out of phase.

1.2 Temporal MMD for QT Interval Preservation

A known limitation of standard autoencoders is the attenuation of low-amplitude signal terminations, notably the T-wave offset. The QT interval tracks the cycle of ventricular depolarization and repolarization. The baseline MSE model struggles with this feature, exhibiting a QT interval absolute error of 138 $\pm$ 15 ms.

Incorporating the Maximum Mean Discrepancy (MMD) with a temporal kernel (Mask `1001014`) reduces this error to 52 $\pm$ 8 ms. MMD acts by aligning the overall temporal distributions of the feature maps, effectively penalizing the model when the reconstructed signal's temporal support shrinks. This strictly preserves the T-wave termination point, yielding a morphological fidelity critical for downstream algorithms that assess conditions like Long QT Syndrome.

1.3 Degradation from Unconstrained Derivative Penalties

Conversely, we observed significant performance degradation when employing the Derivative penalty without spatial regularizers (Mask `1010010`). This configuration caused the overall amplitude MAE to increase to 67.3 μV, and the QT interval error to reach 402 ms.

The mechanism behind this failure is evident: the Derivative loss penalizes mismatches in local slope. When the model experiences even a minor temporal phase shift, it attempts to match the target slope prematurely, causing amplitude overshoots. This destabilizes the low-frequency baseline scaling and distorts the overall complex boundaries.

1.4 Energy Difference for Variance Preservation

Finally, the Energy Difference (ED) penalty proved highly effective at mitigating variance shrinkage. Models utilizing the ED penalty consistently maintained a variance ratio near 1.0 (e.g., Mask `1001113` achieved a log variance ratio error of 0.73 $\pm$ 0.05). This ensures that the reconstructed QRS amplitudes maintain their true clinical magnitude, preventing the signal attenuation typically induced by standard MSE losses.